Experimental Setup and Methodology

Vibration and acoustic health monitoring of a PMSM-driven four-wheeler electric vehicle

1. Experimental Setup

The work is done on a four-wheeler electric vehicle whose traction motor is a permanent magnet synchronous motor (PMSM). Two signals are recorded together while the motor runs: the sound it radiates and the vibration of its body. Sound tells us how the motor is perceived, and vibration tells us what is happening on the motor itself, so using both should give a more reliable picture of its condition.

1.1 Test vehicle and motor

The motor is fed by an inverter from the battery pack. A few features of a PMSM matter for this work.

- The magnets sit in the rotor and the back-EMF is sinusoidal, so the motor is normally run with field oriented control.
- The motor drives the wheels through a reduction gear, so gear and bearing noise also reach the sensors.
- The inverter switches at high frequency, which adds a tonal whine on top of the normal motor sound.

The nameplate data of the motor (rated power, rated speed, number of poles) is noted before the tests, because the pole and slot numbers are needed later to locate the harmonics in the vibration spectrum.

1.2 Noise and vibration sources

Source	How it shows up	Seen mainly by
Electromagnetic	Torque ripple, cogging and slot harmonics at multiples of the electrical frequency	Accelerometer and microphone
Inverter switching	A high pitched tonal whine at the switching frequency and its sidebands	Microphone
Mechanical	Bearing defects, rotor imbalance, misalignment and gear mesh	Accelerometer
Aerodynamic and cooling	Broadband noise that rises with speed	Microphone

1.3 Sensors

All the sensors are MEMS (Micro-Electro-Mechanical Systems) devices. They are small, light, cheap and need very little power, so they can be fitted in the tight space around a motor in a car.

Signal	Sensor	Use
Sound	MEMS microphone	Records the noise from the motor, from which the psychoacoustic features are calculated.
Vibration	Tri-axial MEMS accelerometer (ADXL series)	Measures the acceleration of the motor housing in three directions.
Sound (extra)	Mobile phone microphone	A low cost option for recording sound only. Phones are not used for vibration.

Placement. The accelerometer is fixed firmly on the motor housing close to the bearing end, and a second one can be placed on the motor mounting bracket to see how much vibration passes into the body. The microphone is held at one fixed distance from the motor. The same positions are used in every run so that the results can be compared. The motor speed is also logged along with the recordings, because the harmonics of a PMSM move with speed.

1.4 Data acquisition hardware

The MEMS sensors are read by a microcontroller (Arduino). The data then goes to a Raspberry Pi, a single-board computer, which stores the recordings, does the processing and carries the trained machine learning model. An IoT part is planned on top, so the motor condition can be seen remotely on a phone or a dashboard.

The path is: motor $\rightarrow$ MEMS sensors $\rightarrow$ microcontroller $\rightarrow$ Raspberry Pi (processing and ML model) $\rightarrow$ IoT output.

1.5 Operating conditions

Tests are planned with the vehicle on a safe stand or on a dynamometer, with the drive wheels off the ground or loaded in a controlled way, so that speed and load can be set and repeated.

Factor	Conditions
Motor health	Healthy motor, and faulty motor with a fault that can be created safely, for example an unbalanced phase or a bearing defect
Speed and load	Steady low, medium and high speed, and a loaded condition
Environment	Without background noise and with background noise (recorded road and traffic noise played during the run)

Background noise is included on purpose. A car is never tested in a silent room in real life, so the method has to work with traffic and road noise around.

2. Methodology

2.1 Workflow

1. Set the vehicle in one operating condition.
2. Record sound and vibration together, along with the motor speed.
3. Cut the recordings into short windows of equal length.

4. Calculate the acoustic and the vibration features for each window.
5. Put them in one labelled matrix.
6. Divide the data into training, validation and testing sets.
7. Train the machine learning models.
8. Check the results with accuracy and confusion matrices.
9. Repeat with background noise.

2.2 Recording and windowing

Sound and vibration are cut into windows in the same way, so every window has a sound part and a vibration part that belong together. A short window with a high overlap is used. This gives a large number of samples from one recording and still keeps the fast changes caused by a fault. The sampling rate of the accelerometer is chosen high enough to cover the slot harmonics and the switching related range we want to look at.

2.3 Acoustic features

Instead of using only the sound pressure level, the sound is described the way a person hears it. Four psychoacoustic features are calculated for each window.

- **Loudness** – how strong the sound seems.
- **Sharpness** – how much high frequency content it has.
- **Fluctuation strength** – slow changes in the sound.
- **Roughness** – fast changes in the sound.

Because the inverter of a PMSM produces a clear whine, tonality is added as one more feature. It measures how tonal a sound is, which is what a switching whine or a gear whine looks like.

2.4 Vibration features

- **Time domain:** RMS (overall vibration energy), peak value, crest factor and kurtosis. Kurtosis rises when there are impacts, such as from a damaged bearing.
- **Frequency domain:** an FFT of each window gives the dominant frequency and the energy in selected bands.
- **Order based:** since the speed is known, the energy is also read at multiples of the rotation and electrical frequency. A change at these orders points to imbalance, an electrical fault or a magnet problem.
- **Axes:** features are taken for each of the three axes and for the resultant.

2.5 Dataset

Each window gives one row of data. The columns are the acoustic features followed by the vibration features, and the row carries the label of the condition. Around 1000 windows are taken for each class, so every class gives a matrix of 1000 rows. Three versions of the data are prepared: acoustic features only, vibration features only, and the two joined together (sensor fusion).

Purpose	Share of data
Training	70%
Validation	15%
Testing	15%

2.6 Machine learning models

Three models are used: an Artificial Neural Network (ANN), a Support Vector Machine (SVM) and a 1D Convolutional Neural Network (1D-CNN). They are trained on the same data so they can be compared fairly. Each model is trained three times, once on each version of the data, which shows whether vibration adds anything to the sound.

2.7 Evaluation

The models are judged by accuracy and by confusion matrices, which show which conditions get mixed up. Everything is done without and with background noise. The main questions are whether the fused data does better than sound alone, and how much the accuracy falls when noise is added.

2.8 Deployment

After training, the best model is put on the Raspberry Pi. Live data from the MEMS sensors passes through the microcontroller, the same features are calculated, and the model reports the condition of the motor. The result can then be sent through the IoT part for remote monitoring of the vehicle.